

Phase	Purpose	Tools
CI/CD for ML	The continuous integration/continuous delivery tool chain	Continuous Machine Learning (CML)
Data exploration	Tools to perform the data exploration	Apache Zeppelin, Google Colab, Jupyter Notebook, JupyterLab, JupyterText, Polynote, etc
Data management	Tools for performing the data management phase	Arrikto, Data Version Control(DVC), and Intake.
Data processing	Tools for data processing and pipelining	Airflow, Hadoop, and Spark
Data visualisation	Visualisers for the data	Dash, Data Studio, Metabase, Redash, Superset, Tableau, etc
Feature store	Feature store tool for data serving	Butterfree and Feast
Hyper parameter tuning	Tools for the ML/DL model hyperparameter tuning	Hyperas, Hyperopt, Katib, Optuna, Scikit Optimize, Tune, etc
Knowledge sharing	Tools for knowledge collaboration	Knowledge Repo, Kyso, etc
ML platforms	Complete ML platform	Algorithmia, Allegro AI, CNVRG, Cubonacci, DAG-sHub, Dataiku, DataRobot, Domino, Gradient, H2O, Hopsworx, Iguazio, Knime, Kubeflow, LynxKite, ML Workspace, Modzy, Pachyderm, Polyaxon, Sagemaker, and Valohai
Model life cycle	Tools to manage the ML model life cycle	Comet, Mlflow, ModelDB, Neptune AI, Sacred, etc
Model serving	Tools for deploying and serving the production model	BentoML, Cortex, GraphPipe, KFServing, PredictionIO, Seldon, Streamlit, TensorFlow Serving, TorchServe
Model optimisation	To manage model scalability	Dask, DeepSpeed, Fiber, Horovod, Mahout, MLlib, Modin, Petastorm, Rapids, Ray, Singa, Tpot, etc
Simplification tools	Tools related to ML simplification and standardisation	Hermione, Koalas, Ludwig, PyCaret, and Turi Create
Workflow tools	Tools pertaining to create ML pipelines	Argo, Flyte, Kale, Kedro, Luigi, Metaflow, Prefect, etc

Table 1: Taxonomy of the MLOps tools [Ref: <https://github.com/kelvins/awesome-mlops>]

Taxonomy

We have described the basic idea of MLOps above. We can categorise the various tools available to us based on the taxonomy given in Table 1 above.

To conclude, MLOps is the efficient and holistic management of ML pipelines and models from an end-to-end perspective, whereas DevOps talks about the management of code and bringing development closer to the operation. This article discussed some of the challenges related to ML technical debt management, and then presented a comprehensive taxonomy of tools across the whole spectrum of MLOps. Of course, as this is an emerging field, tools are being added every other day. **END** 

References

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